Search or **Learning?**

The 2012 Big Bang that redefined machine intelligence and sparked the dawn of modern Deep Learning.

Three Pillars That Made AlexNet Possible

Hardware Revolution

Moving away from CPUs, models were trained on modified consumer gaming graphics cards (GPUs), unlocking massive parallel processing power.

Data Abundance

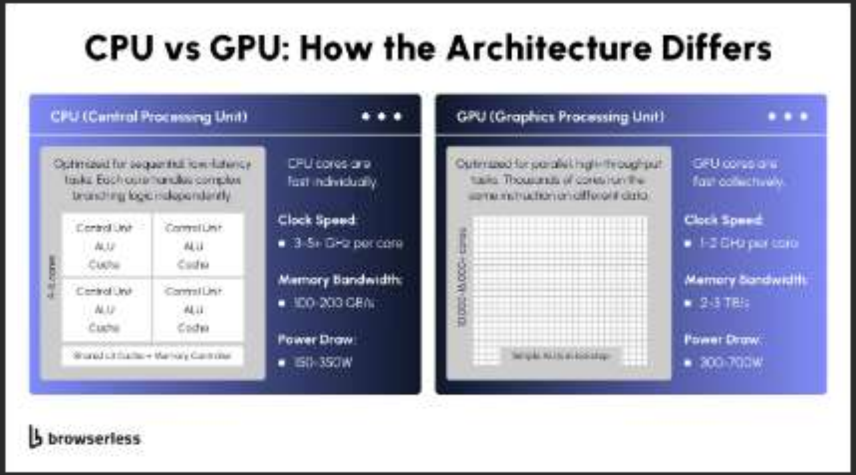
To prevent the model from memorizing, it needed vast fuel. The ImageNet project provided this with 1.2 million cleanly labeled images.

Algorithm Innovation

Mathematical bottlenecks in deep networks were solved using the ReLU activation function, and overfitting was curbed using the Dropout technique.

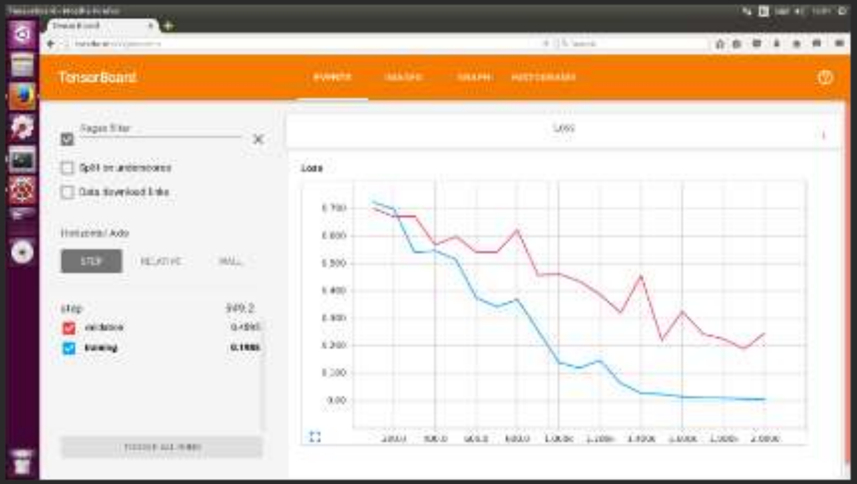
The Processing War: CPU vs. GPU

- **CPU (Central Processing Unit):** Few, but highly capable cores. Like an expert math professor solving complex logic sequentially. Cannot perform millions of simple tasks simultaneously.
- **GPU (Graphics Processing Unit):** Thousands of small, simple cores. Like thousands of students doing basic math at the same time. Perfect for matrix multiplications in deep learning.
- **The Time Breakthrough:** Training AlexNet's 60M parameters on a CPU would take months. Using dual GPUs reduced this to 5-6 days, accelerating the pace of AI innovation.



Visualizing Learning: TensorBoard

- **Adding Layers (Depth):** As networks deepen, they learn increasingly complex features—from straight edges in early layers to complex textures in later ones.
- **The Loss Curve:** Represents the model's error rate. During training, we expect this curve to steadily decrease towards zero.
- **The Danger - Overfitting:** If Training Loss continues to drop, but Validation Loss (error on unseen data) sharply spikes, the model has stopped generalizing and is merely memorizing the data.



The Evolution of "Intelligence"

Feature	Deep Blue (1997)	AlexNet (2012)
Paradigm	Symbolic AI (Expert Systems)	Connectionist AI (Machine Learning)
How It Works	Searches through rules written by human grandmasters at immense speeds (brute-force).	Discovers complex patterns in pixels **independently**, without human-coded rules.
Perception of Intelligence	We taught it chess. It is a super-fast calculator unable to step outside its programmed rules.	It **"generalizes"** from data. This marks the transition from coded logic to self-building systems.

Thanks For Listening

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